A Semi-physical Model of Closed-loop IFOG and Ways to Improve the Dynamic Performance

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Abstract

Dynamic performance is an important indicator of fiber optic gyroscopes, indicating the gyroscopes tracking ability in the state of large angular acceleration, such as overload, swaying, vibration. The Interferometric Fiber Optic Gyroscopes(IFOG) is based on the Sagnac effect and has the characteristics of nonlinearity, input saturation and time delay, etc. In general the IFOG’s closed-loop system is approximated as a linear system for modeling and analysis. This linear model is just suitable for the IFOG working in a static or uniform motion. In high dynamic conditions, the IFOG tracking error will appear strong nonlinearity and even become zero， which cannot be simply treated as linear model. Based on the analysis of the high dynamic tracking error characteristics of the IFOG, a semi-physical simulation method is proposed which can simulate the dynamic working environment of the IFOG under laboratory environment conditions. By equivalent external excitation to $-\pi \sim +\pi$ phase, an arbitrary angular acceleration input is achieved. Comparing the external excitation and the IFOG output, the IFOG's Dynamic response capability is truly reflected. Simulation show that it is beneficial in three aspects for improving the IFOG's tracking ability. Firstly, reduce the delay time which is produced by data processing in the processor, Improve the rapidity of the IFOG's response; Secondly, A dynamic modulation depth which is proportional to the tracking error technology can be implemented. Thirdly, the system feedback gain is combined of a optimal static gain and a optimal dynamic gain, which is decided by the tracking error.

1. Introduction

The Closed-loop IFOG is one of the most successful commercial fiber sensors [1]. Compared with traditional mechanical gyroscope, the fiber optic gyroscope has more advantages, such as, full solidity, a larger dynamic range, lower cost, lower power consumption, smaller in size, light in quality, etc. Nowadays, it has been widely used in strapdown inertial navigation system (SINS). In the performance index system of IFOG, dynamic performance is an important indicator of IFOG, indicating the gyroscopes tracking ability in the state of large angular acceleration, such as overload, swaying, vibration [2]. Due to the inherent nonlinearity and periodicity of IFOG, the saturation characteristics of the A/D converter, the truncation error and time delay in the digital processing, etc. these have greatly affected the dynamic performance of the IFOG [3-7].

It is difficult to build complete mathematical modeling of fiber optic gyroscope directly. Junliang Han has developed a simplified mathematical model of closed-loop IFOG, which was obtained by linearizing the interference fringes [3]. This is also a common method for analyzing the closed-loop system of Closed-loop IFOG. But this linear model is just suitable for the IFOG working in a static or uniform motion. In high dynamic conditions, the IFOG tracking error will appear strong nonlinearity and even become zero， which cannot be simply treated as linear model.

In this paper, we develop a semi-physical simulation model of Closed-loop IFOG. By equating the rotational rate excitation to a phase modulation signal, and applying it to the integrated optic chip (MIOC), an arbitrary external angular velocity simulation is achieved. The bandwidth test was carried out to verify the correctness of the model, which were consistent with the IFOG test results on the angular vibration table. Using this dynamic simulation model, we have conducted dynamic performance experimental on the three parameters of time delay, feedback proportional coefficient, and modulation depth in the IFOG. The experimental results and the improved method are given, which provides a certain reference for the research of optimizing the dynamic characteristics of gyroscope.

2. Principle and analysis

2.1. Basic Principle of closed-loop IFOGs

Fiber optic gyroscope is based on the Sagnac effect and made in co-called minimum configuration with digital signal processing [1-2]. Minimum configuration ensures the reciprocity of optical paths for two beams counter-propagating in a fiber loop. Structural scheme of the closed-loop IFOG is represented in Figure 1.

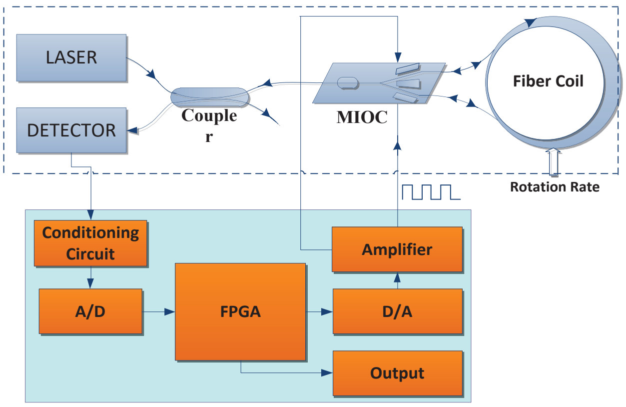

Figure 1. Structural scheme of digital closed-loop IFOG.

The light from the laser is divided into two counterpropagating light waves in the fiber coil, and the phase difference of the pair of light waves is proportional to the rotation rate of the closed optical path. The intensity of interference light, carrying the rotation rate information, is converted into a voltage signal by photodetector, which is then amplified and filtered by the signal conditioning circuit. The signal is demodulated and integrated twice in the digital processor to generate a phase ramp which is applied to the MIOC. The height of the phase ramp step $\phi_F$is equal to the Sagnac phase shift ϕ_s, The total phase difference $\Delta\phi = \phi_s - \phi_F$ is servo-controlled on zero. For maximum sensitivity, a square wave modulation ϕ_b, usually $\pm\pi/2$ rad, is produced and added to the phase ramp [4]. The block diagram of closed-loop IFOG can be described in Figure 2.

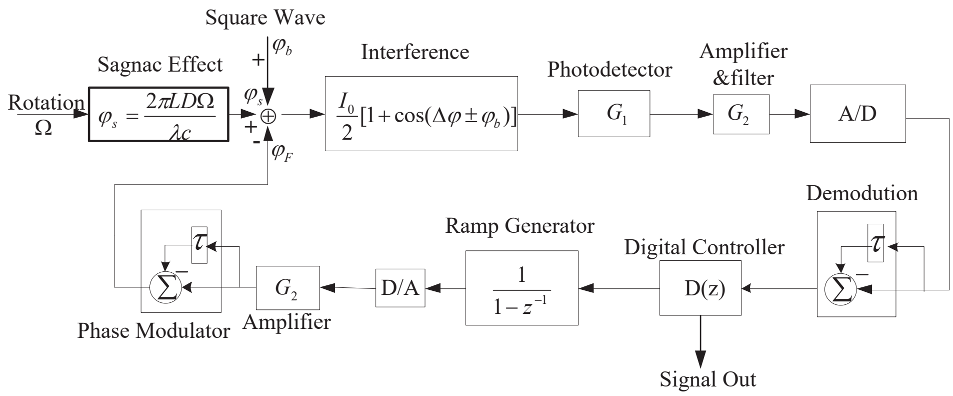

Figure 2. Block diagram of digital closed-loop IFOG configuration.

2.2. Analysis of IFOG's closed-loop system

The closed-loop of IFOG is a strongly nonlinear system, and its steady-state performance and dynamic performance usually restrict each other. The nonlinearity of the IFOG is determined by two aspects: the first is determined by the gyroscope demodulated error in the form of a cosine function; and the second is the saturation characteristic of the A/D converter.

Nonlinearity and periodicity of the IFOG demodulated error. When there is a rotation rate tracking error, the photodetector of the gyro outputs a square wave signal with the same frequency as the modulation square wave ϕ_b, as shown in Figure 3. Sampling the voltage on the adjacent period of the detector and making a difference, we can get the tracking error, which can be written:

$$\begin{aligned}\Delta V(\phi_s,\phi_0) &= kP_0[\cos(\phi_s-\phi_F+\phi_b)-\cos(\phi_s-\phi_F-\phi_b)] \\ &= kP_0\sin\phi_b\sin(\phi_s-\phi_F)\end{aligned} \quad (1)$$

Where: K is the photoelectric conversion factor and amplification factor.

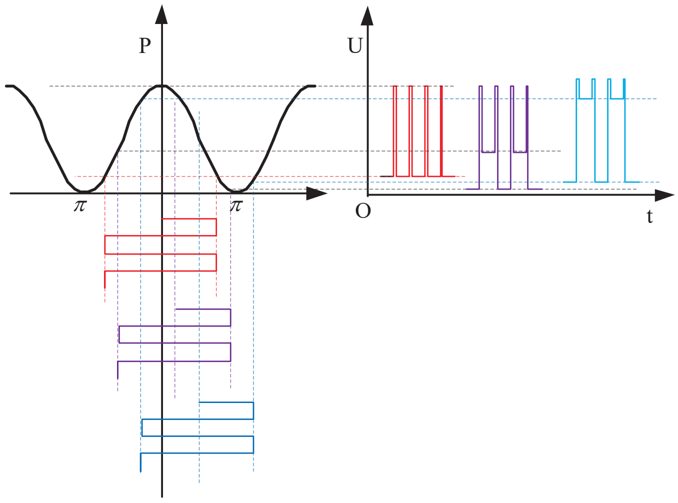

Figure 3. Square-wave biasing modulation and demodulation.

The saturation of the error response because of the limited sampling range of the A/D converter. In order to improve the sampling resolution, the A/D converter input range is usually less than ±2V. It is possible that the signal exceeds the A/D converter positive and negative sampling range with the square wave modulation$\phi_b = \pm\pi/2$ rad, and the error response becomes a fixed value when there is a big racking error, as shown in the figure 4. In this case the error response becomes:

$$\Delta I(\phi_s,\phi_0)=\begin{cases} kP_0\sin\phi_0\sin(\phi_s-\phi_F) & \text{no saturated} \\ V\_AD_{max}-V\_AD_{min} & \text{all saturated} \end{cases} \quad (2)$$

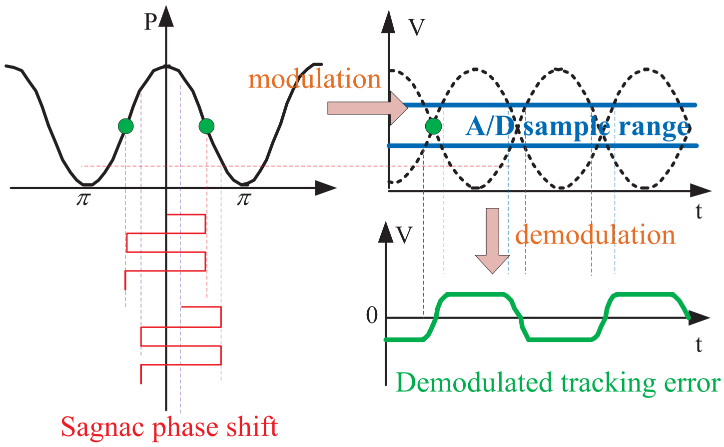

Figure 4. The saturation of A/D converter with modulation depth $\phi_b = \pm\pi/2$.

The failure of the error response because of over-modulation of square wave. In order to suppress the relative intensity noise (RIN) brought by the high-power light source [8], the over-modulation technology, which the depth of square-wave modulation ϕ_b is may be chosen between $\pi/2$ and π instead of $\pi/2$, usually implemented. In this way, the working point of the gyro will be at the bottom area of the interference fringe instead of the middle, as shown in the figure 5. When there is tracking error, there will be three situations in the sampling of A/D converter: The first is that when the tracking error is small, the A/D converter sampling voltage is not saturated, and the formula 1 is applicable; The second is that when the tracking error increases, one of the square wave modulation states occurs positive saturation, the other is still within the sampling range. In this case, the error response becomes decrease with the tracking error increases; the third is both of the square wave modulation states occurs positive saturation when the tracking error increases further, and the difference of voltage on the detector becomes zero, as shown in the figure 5. It means that the IFOG loses the ability to respond to the tracking error until the sampling voltages exit the situation of positive saturation. So, the error response becomes:

$$\Delta I(\phi_s,\phi_0)=\begin{cases} kP_0\sin\phi_0\sin(\phi_s-\phi_F) & \text{no saturated} \\ V\_AD_{max}-kP_0\cos(\phi_s-\phi_F-\phi_b) & \text{half saturated} \\ V\_AD_{max}-V\_AD_{max}=0 & \text{all saturated} \end{cases} \quad (3)$$

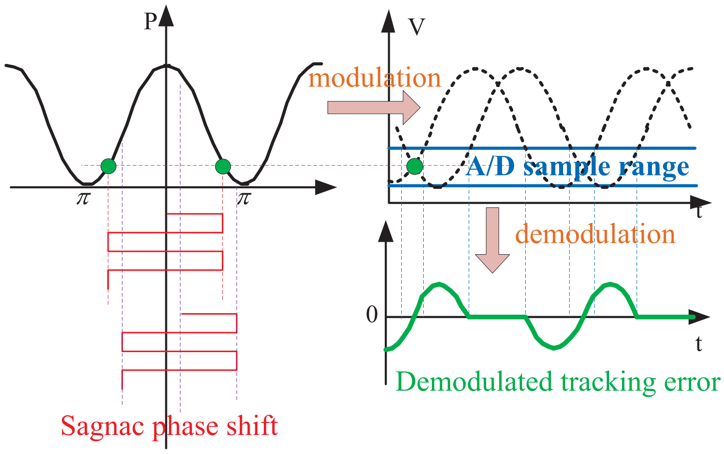

Figure 5. The saturation of A/D converter with over-modulation.

Pure time delay in the gyro closed loop system. The central processor in IFOG is usually a Field Programmable Gate Array (FPGA). Its data processing is based on the transit time of the fiber coil. Pure time delay which is proportional to the length of the fiber coil is produced in the process of demodulation, integration, accumulating phase ramp and square wave modulation, etc. When there is a rotate rate change, the gyro will not have feedback until the delay time is over. The longer the pure delay time, the more likely the gyro will oscillate, It will cause deterioration of the dynamic performance of IFOG.

In addition to the above analysis, there are still data truncation errors in FPGA, quantization errors in A/D and D/A converter, cross coupling between rotation rate closed loop and feedback channel gain closed loop [9]. In summary, IFOG is a complex control system with nonlinear response, input saturation, cross-coupling, truncation error, etc. It will be very difficult and unrealistic to directly build a mathematical model. Therefore, a simulation system that truly reflects the dynamic performance of the IFOG is necessary.

3. A Semi-physical Model of Closed-loop IFOG

3.1. Model of Closed-loop IFOG

We developed a semi-physical simulation system which can simulate the dynamic working environment of the IFOG under laboratory environment conditions. The semi-physical simulation system is divided into two parts. One is the excitation signal which is applied to the MIOC to simulate arbitrary angular velocity input; the other is the closed loop system of the IFOG to respond to the excitation signal.

The excitation signal generation, an arbitrary angular velocity corresponding to Sagnac phase ϕ_s can be described as:

$$\phi_s = \frac{2\pi LD}{\lambda c}\Omega \tag{4}$$

Where: L is the coil length, D is the coil average diameter, λ is the average source wavelength, and c is the light speed in vacuum.

The excitation signal is applied to the MIOC and the resulting phase shift ϕ_e is [8]:

$$\phi_e = K_m[V_m(t) - V_m(t-\tau)] \tag{5}$$

Where: K is the electro-optic modulation of MIOC.

Because of the excitation signal applied to the MIOC cannot rise indefinitely, the phase shift ϕ_e generated by the excitation signal can be equivalent to $-\pi \sim +\pi$ phase by using the periodicity of the interference fringe:

$$-\pi \le \phi_{jili} = \phi_s \pm 2n\pi < \pi \tag{6}$$

Taking constant angular acceleration α as an example, the excitation signal can be generated as follows:

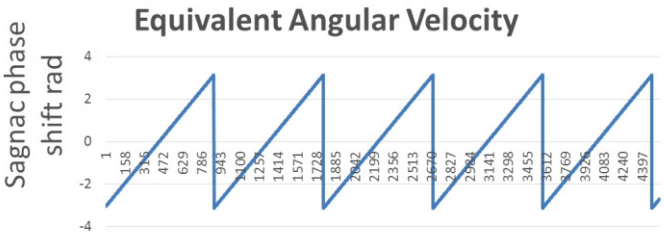

Figure 6. Equivalent angular velocity.

Form formulas (4),(5) and (6), the excitation signal is then:

$$V_m(t) = \frac{2\pi LD}{K_m \lambda c}\Omega \pm \frac{2n\pi}{K_m} + V_m(t-\tau) \tag{7}$$

Put the equivalent angular velocity data in Figure 5 into the formulas (7), we can get the finally excitation signal applied to MIOC as follows:

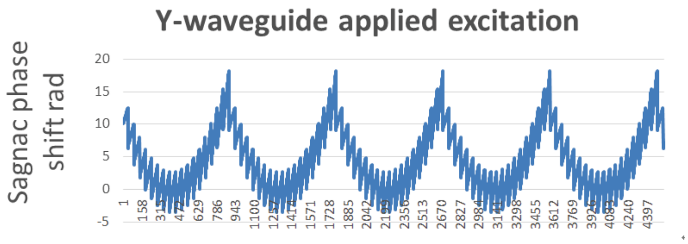

Figure 7. The excitation signal applied to MIOC.

In addition, In order to generate the excitation signal, the following aspects should also be noted:

1 The frequency of the excitation signal should be much Higher than the eigen frequency of the IFOG, such an more than 10 times;

2、The 2π reset of the excitation signal should be synchronized with the square wave modulation signal to avoid the gyroscope data fluctuation during the sampling period;

3、After each 2π reset of the excitation signal, a blank segment without angular rate information should be inserted, which is removed during demodulation.

3.2. The Semi-physical Model correctness verification

Take the fiber optic gyroscope with diameter of 120mm as an example for simulation verification. The gyroscope parameters are as follows.

Table 1. The gyroscope parameters

parameter	**value**	**unit**
optical power	3	mW
Light wavelength	1550	nm
detector conversion efficiency	0.02	V/μW
Fiber coil length	2000	m
Fiber coil diameter	110	mm
Circuit magnification	10	/
A/D input range	-1~+1	V
Modulation depth	$7\pi/8$	

Input sinusoidal signal as excitation to verify the correctness of the model. The amplitude peak value of the input sine signal is 30°/s, and the frequency is from 50Hz to 300Hz. When the input angular frequency reaches 200Hz, there is a small amount of saturation in the A/D converter, but the gyroscope's tracking performance is good at this time, as shown in the figure 8; When the input angular frequency exceeds 300Hz, the saturation of the A/D converter increases, and even the situation of continuous saturation in multiple transit times appears (this phenomenon needs to be avoided). At this time, the tracking ability of the gyro decreases, and even fails, as shown in the figure 9. It is consistent with the actual gyroscope angular vibration test, which verifies the correctness of the model.

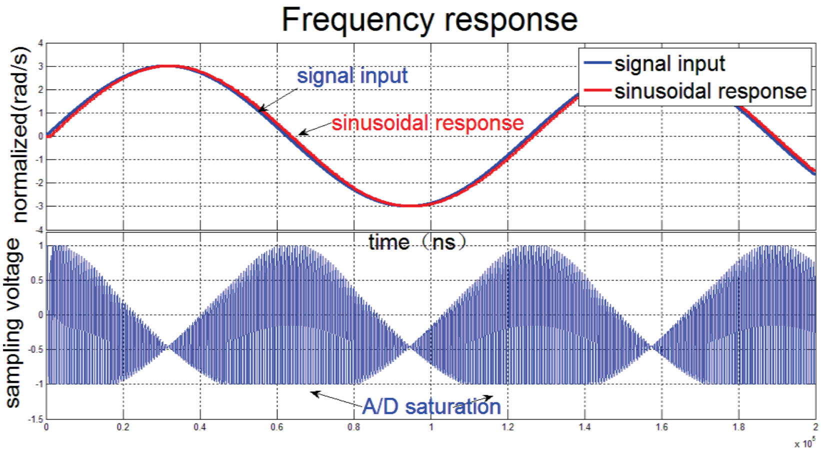

Figure 8. Tracking performance with angular frequency 200Hz.

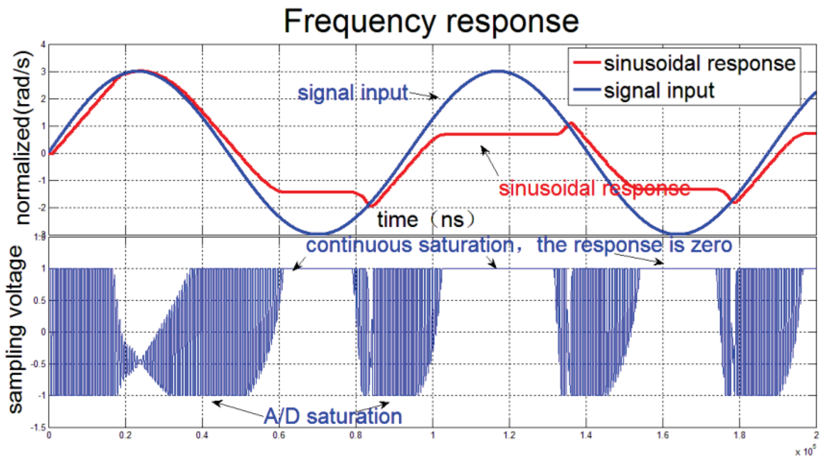

Figure 9. Tracking performance with angular frequency 300Hz.

4. Improving the closed-loop IFOG's tracking ability

Analysis and Simulation show that it is beneficial in three aspects for improving the IFOG's Dynamic performance.

4.1. The Influence of Delay Time on Step Response

Apply a step signal which amplitude is 5°/s to the simulation system, and compare the step response of the gyro with different delay times. The test results are consistent with theoretical analysis: the longer the delay time, the greater the overshoot of the gyro's step response and the easier it is to oscillate, as show in the figure 10. In addition, it can be obtained that in a high delay time of IFOG, in order to reduce the overshoot, the feedback proportional coefficient of the system must be sacrificed, which will increase the rise time of the gyroscope's step response and damage the dynamic performance of the gyroscope.

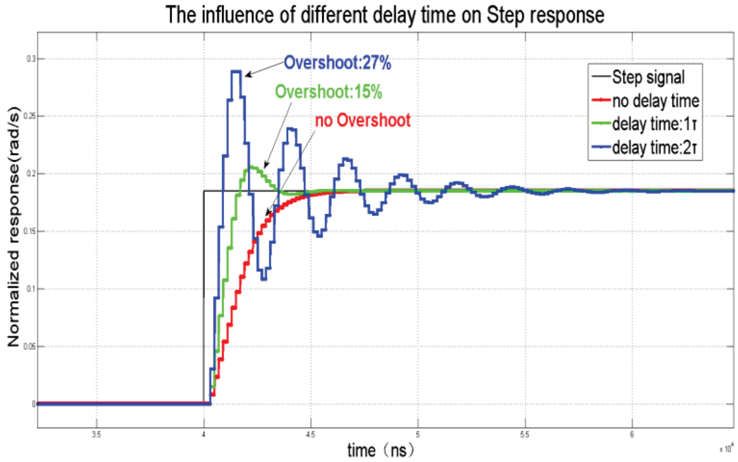

Figure 10. Step response with different Delay Time.

In the gyroscope design, the delay time of the system must be minimized. For a gyroscope with a long fiber coil, it is also necessary to consider multiple closed loop feedback in one transit time to make the system delay time less than one transit time.

4.2. The Influence of Modulation Depth on Frequency Response

Overmodulation can suppress the relative intensity noise of the light source, but will cause the asymmetry of the A/D converter sampling. When a large tracking error occurs, the demodulation error will be saturated or even invalid.

The sine angular frequency test is performed on $\pm\pi/2$、$\pm\pi 4/5$、$\pm\pi 7/8$ modulation depth respectively, the amplitude peak value is set to 30°/s, and the angular frequency is 250Hz. The result is shown in Figure 11. As the modulation depth increases, the dynamic performance of the gyro gradually decreases. The reasons is that the greater the modulation depth, the more likely it is to overflow in the same direction when tracking errors occur.

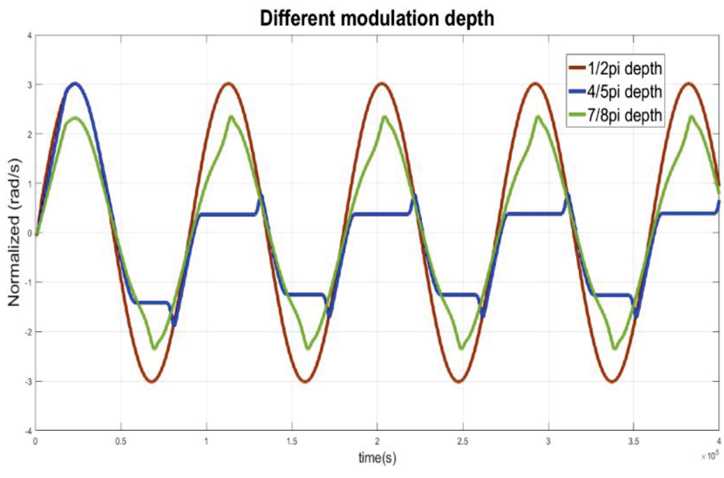

Figure 11. Frequency response with different Modulation Depth.

A dynamic modulation depth technology can be implemented. In the static state, a deeper modulation depth is used; when the dynamic tracking error increases, the modulation depth should be reduced.

4.3. The Influence of Feedback Proportional Coefficient on Slope Response

According to the simplified closed-loop transfer function of the closed-loop IFOG, the gyro closed-loop system is a first-order system, and there is a constant tracking error for the ramp input signal, and the error size is inversely proportional to the feedback proportional coefficient of the IFOG. However, due to the saturation of the A/D converter in the gyroscope, the output of the gyroscope may diverge, that is, the IFOG has a limited angular acceleration tracking ability.

Input a slope ramp signal with an acceleration of 20000°/s, and compare the tracking conditions with feedback proportional coefficient. The experimental results show that: the larger the feedback coefficient in the gyro corresponds to the better dynamic tracking ability, but it may cause the gyro to oscillate; the smaller feedback coefficient in the gyro can reduce steady-state oscillation, however, when tracking the slope signal, the closed-loop error continues to increase, and the A/D converter has continuous positive saturation, resulting in the system's final tracking failure, as shown in Figure 12.

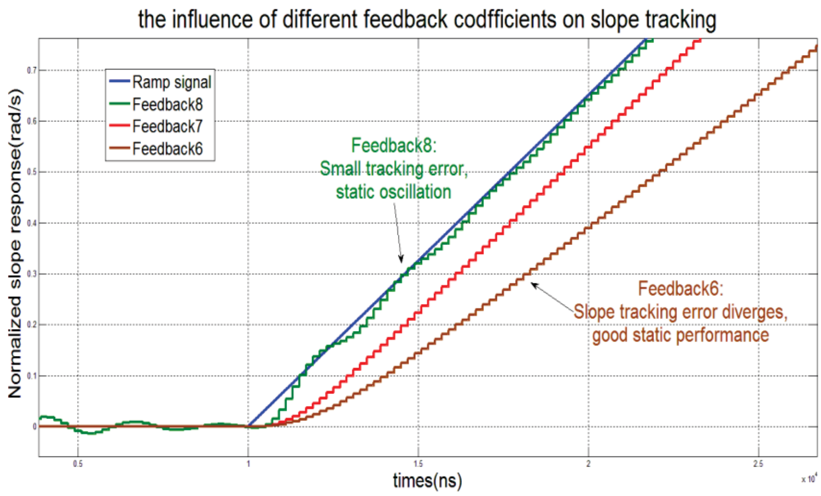

Figure 12. Slope response with different Feedback Proportional Coefficient.

Similarly, In the gyroscope design, the system feedback proportional coefficient is combined of a optimal static coefficient and a optimal dynamic coefficient, which is decided by the tracking error. The general principle is that the bigger the tracking error, the bigger the feedback proportional coefficient should be.

5. Conclusion

In this paper, through the analysis of the dynamic performance of the closed-loop IFOG, we get that the gyro is a complex closed-loop system with strong nonlinearity, pure time delay, truncation error, etc. For further study, we have devopled a semi-physical simulation method of IFOG, and verified the correctness of the simulation system. Finally, we conducted experiments with different delay time, modulation depth, feedback proportional coefficient , and based on the experimental results, we gave some methods to improve dynamic performance.

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